

2. Poisson brackets: Determine: (a) the Poisson brackets formed from the Cartesian components of the momentum $\mathbf{p}$ and the angular momentum $\mathbf{M} = \mathbf{r} \times \mathbf{p}$ of a particle, and (b) the Poisson brackets formed from the Cartesian components of the angular momentum $\mathbf{M}$.

$$\{A, B\} = \sum \frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial B}{\partial q_i} \frac{\partial A}{\partial p_i}$$

$$\begin{aligned} a) \{p_x, M_x\} &= \cancel{\frac{\partial p_x}{\partial x} \frac{\partial M_x}{\partial p_x}} - \cancel{\frac{\partial M_x}{\partial x} \frac{\partial p_x}{\partial p_x}} = 0 \\ &+ \cancel{\frac{\partial p_x}{\partial y} \frac{\partial M_x}{\partial p_y}} - \cancel{\frac{\partial M_x}{\partial y} \frac{\partial p_x}{\partial p_y}} \\ &+ \cancel{\frac{\partial p_x}{\partial z} \frac{\partial M_x}{\partial p_z}} - \cancel{\frac{\partial M_x}{\partial z} \frac{\partial p_x}{\partial p_z}} \end{aligned}$$

$$\begin{aligned} \{p_x, M_y\} &= -\frac{\partial M_y}{\partial x} \\ &= p_z \end{aligned}$$

$$\begin{aligned} M_y &= -(x p_z - z p_x) \\ M_y &= z p_x - x p_z \end{aligned}$$

$$\{p_x, M_z\} = -\frac{\partial M_z}{\partial x} = -p_y$$

$$M_z = x p_y - y p_x$$

$$\begin{aligned} b) \{M_x, M_y\} &= \cancel{\frac{\partial M_x}{\partial x} \frac{\partial M_y}{\partial p_x}} - \cancel{\frac{\partial M_y}{\partial x} \frac{\partial M_x}{\partial p_x}} \\ &+ \cancel{\frac{\partial M_x}{\partial y} \frac{\partial M_y}{\partial p_y}} - \cancel{\frac{\partial M_y}{\partial y} \frac{\partial M_x}{\partial p_y}} \\ &+ \frac{\partial M_x}{\partial z} \frac{\partial M_y}{\partial p_z} - \frac{\partial M_y}{\partial z} \frac{\partial M_x}{\partial p_z} \end{aligned}$$

$$\begin{aligned} M_x &= y p_z - z p_y \\ M_y &= z p_x - x p_z \end{aligned}$$

$$\begin{aligned} &= (-p_y)(-x) - (p_x)(y) \\ &= x p_y - y p_x \\ &= M_z \end{aligned}$$

3. More Poisson brackets: Show that: (a) $[\Phi, M_z] = 0$, where Φ is any function, spherically symmetric about the origin, of the coordinates and the momentum of a particle, and (b) $[\mathbf{f}, M_z] = \mathbf{f} \times \hat{z}$, where $\mathbf{f}$ is a vector function of the coordinates and the momentum of a particle and $\hat{z}$ is the unit vector along the positive z -axis.

$$a) \quad \Phi \equiv \Phi(r^2, p^2, \vec{r} \cdot \vec{p})$$